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Late Cretaceous paleomagnetism of the Tombstone district and vicinity: Evidence for a rotational domain boundary in southeastern Arizona

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Abstract. Paleomagnetic data are presented for Late Cretaceous volcanic and intrusive rocks in the Tombstone district and Dos Cabezas Mountains of southeastern Arizona. The resultant paleopole (73°N, 224°E, A_{95}= 8°) is similar to a Late Cretaceous reference pole for North America (82°N, 210°E, A_{95}= 7°), implying that these rocks have not been significantly rotated about vertical or near-vertical axes relative to the continental interior. Generally, concordant results have also been published for Pennsylvanian terra rosa paleosols within the Black Prince Limestone at localities in the nearby Gunnison Hills, Whetstone Mountains, and Naco Hills. Together, however, these Carboniferous data indicate a 9° ± 7° clockwise rotation, with respect to North America, similar to that determined for the Colorado Plateau (5°-14°) since the mid-Cretaceous. Published paleomagnetic data for rocks south and west of the Tombstone vicinity indicate much larger clockwise rotations ranging from ~25° to ~60°, implying that a rotational domain boundary runs between the Mustang Mountains and Canelo Hills to the southeast and between the Silver Bell and Tucson Mountains to the northwest. This boundary roughly coincides, in part, with the Sawmill Canyon fault zone and is parallel to other northwest trending strike-slip faults in the region that were active during the Laramide orogeny. The clockwise rotations in southern Arizona are likely part of a larger pattern of dextral shear along the western margin of North America during the Late Cretaceous and early Tertiary caused by northward oblique subduction of oceanic plates relative to the stable craton.

Introduction

The driving force of the Laramide orogeny has been attributed to low-angle subduction [*Lipman et al.*, 1971; *Dickinson and Snyder*, 1978], and changes in tectonic patterns within the Cordilleran foreland region were presumably associated with changes in subduction geometry at the western continental margin. Convergence rates between the Farallon and North American plates increased markedly during the Late Cretaceous and early Tertiary, and the relative convergence direction between them also changed [*Engebretson et al.*, 1984; *Stock and Molnar*, 1988]. Rotation of the Colorado Plateau [*Hamilton*, 1981] has been related to a corresponding change in the direction of horizontal compression from east-northeast to more northward [*Chapin and Cather*, 1983; *Gries*, 1983], and such changes would cause contractional or strike-slip deformation on properly oriented structures at different times [*Miller et al.*, 1992]. Local vertical axis rotations are also a likely consequence, and paleomagnetic determinations of them provide valuable information on the sense and distribution of strike-slip deformation in the southern foreland region.

Paleomagnetic data for Late Cretaceous rocks in the Tucson and Silver Bell Mountains of southeastern Arizona (Figure 1) indicate clockwise vertical axis rotations (~10° and 30°, respectively) relative to stable North America [*Hagstrum and Sawyer*, 1989; *Hagstrum and Lipman*, 1991a]. In addition, comparisons of paleomagnetic data for sites within regional Jurassic ash flow sheets show relative rotations of ~30° to 60° between localities in the Mustang Mountains, Canelo Hills, and Huachuca Mountains [*Hagstrum and Lipman*, 1991b]. Paleomagnetic poles for terra rosa paleosols within the Pennsylvanian part of the Black Prince Limestone in the Naco Hills, Whetstone Mountains, and Gunnison Hills, however, are similar to reference Carboniferous paleopoles for North America, implying that these Paleozoic rocks have remained unrotated relative to the stable interior [*Nick et al.*, 1991]. Clarification of this disagreement between data indicating vertical axis rotations and those which apparently do not has important implications for the tectonic history of southeastern Arizona and has particular significance for paleomagnetic data from the region which have been, and in the future might be, used to calculate reference paleopoles for the North American plate [*Vugteveen et al.*, 1981; *Kluth et al.*, 1982; *May et al.*, 1986].

High-quality paleomagnetic data for pre-Cenozoic rocks are difficult to obtain in southern Arizona because of local alteration, multiple structural complexities, and younger sediments which cover more than half the pre-Cenozoic bedrock. Few data points therefore are available with which to discern regional patterns of pre-Tertiary deformation and vertical axis rotation. Moreover, early investigators pointed to the lack of geologic evidence for vertical axis rotations associated with Laramide deformation and Basin and Range faulting in suggesting that southern Arizona has remained stable relative to

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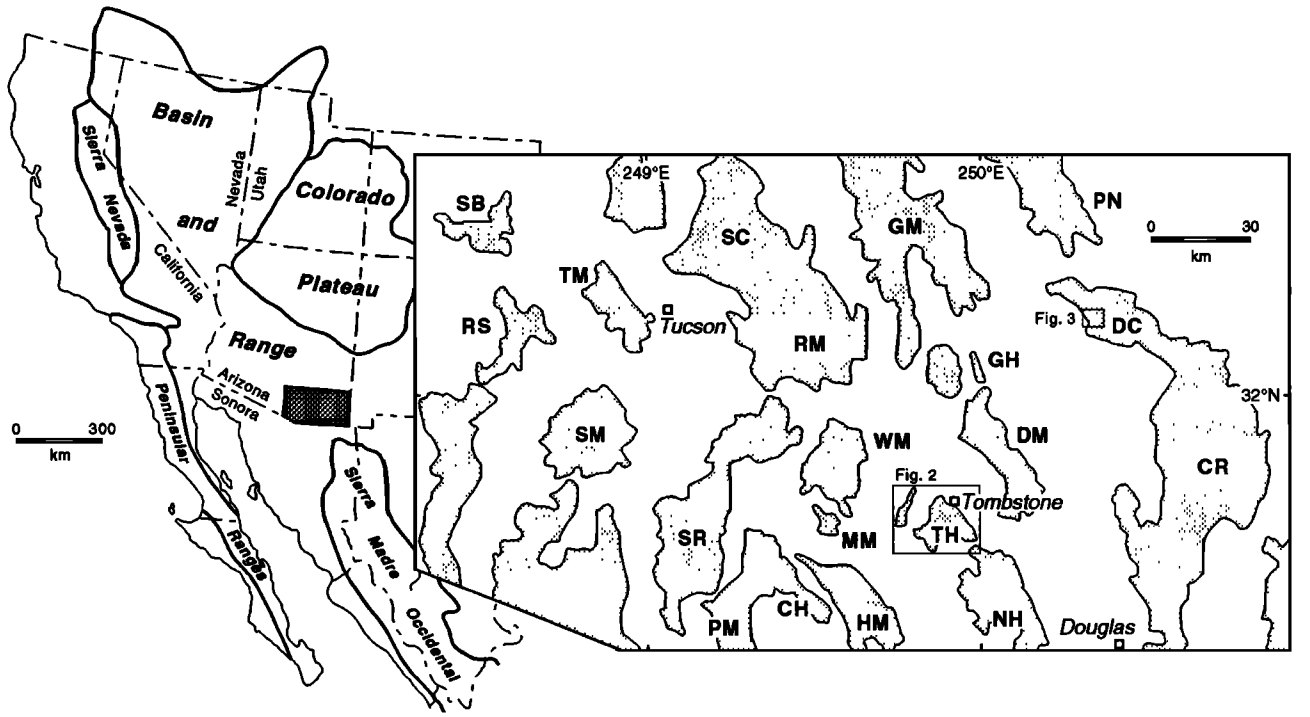


Figure 1. (left) Regional map showing physiographic provinces in southwest North America [after *Stewart*, 1978] and (right) index map showing location of mountain ranges in southeast Arizona (same area as in Figure 9). Areas of Figures 2 and 3 are also indicated. Mountain range abbreviations are CH, Canelo Hills; CR, Chiricahua Mountains; DC, Dos Cabezas Mountains; DM, Dragoon Mountains; GH, Gunnison Hills; GM, Galiuro Mountains; HM, Huachuca Mountains; MM, Mustang Mountains; NH, Naco Hills; PM, Patagonia Mountains; PN, Pinaleno Mountains; RM, Rincon Mountains; RS, Roskruge Mountains; SB, Silver Bell Mountains; SC, Santa Catalina Mountains; SM, Sierrita Mountains; SR, Santa Rita Mountains; TH, Tombstone Hills; TM, Tucson Mountains; and WM, Whetstone Mountains.

North America since the Jurrassic [e.g., *May et al.*, 1986]. In this study we present new paleomagnetic data for Late Cretaceous volcanic and intrusive rocks in the Tombstone Hills and Dos Cabezas Mountains. The paleomagnetic pole for these localities is similar to mid-Cretaceous and Late Cretaceous reference paleopoles for the North American plate, implying that these rocks are also unrotated. Combining the new data with those available for other localities in southeastern Arizona, we propose a simple pattern of rotated and unrotated domains in the region separated by a northwest trending boundary.

Geologic Setting

During Mesozoic time, southeastern Arizona was part of a continental margin volcanic arc that extended along the North American Cordillera from Mexico to Alaska. Jurassic and Cretaceous silicic volcanic rocks are well preserved and only weakly metamorphosed in the region, but they have been structurally dismembered and fragmented due to repeated deformation that includes Laramide compression, middle Tertiary extension, and late Cenozoic block faulting. In southern Arizona the Laramide orogeny has been divided into three phases [e.g., *Miller et al.*, 1992], an initial phase characterized by compressional basement-cored block uplift; a medial phase of major plutonism, porphyry copper mineralization, and regional uplift; and a waning phase of compressional tectonics, plutonism, and volcanism. Northwest trending faults with probable strike-slip displacement are a major tectonic element of southeastern Arizona [*Titley*, 1976; *Drewes*, 1981], and *Keith and Wilt* [1986] have associated left-lateral movement on these faults with the medial phase (65 to 55 Ma) of Laramide deformation. Clockwise vertical axis rotations of Late Cretaceous rocks [e.g., *Hagstrum and Sawyer*, 1989], however, imply dextral shear and right-lateral displacement, at least, during Laramide deformation.

Regional associations between Cretaceous ash flow sheets, caldera sources, and mineral deposits in southern Arizona were reviewed by *Lipman and Sawyer* [1985], and, more recently, Jurassic calderas and their corresponding ash flow sheets have been identified by *Lipman and Hagstrum* [1992]. Two deeply eroded Late Cretaceous caldera fragments in the Tombstone Hills and Dos Cabezas Mountains, discussed by *Lipman and Sawyer* [1985], are the subjects of this paleomagnetic investigation.

In the Tombstone Hills the oldest rocks of Late Cretaceous age are andesitic lava flows and breccias which unconformably overlie a sequence of Precambrian to Cretaceous rocks, the youngest of which are Lower Cretaceous sedimentary rocks of the Bisbee Formation in the southern part of the mining dis-

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trict (Figure 2). These intermediate-composition rocks are likely the earliest exposed eruptive products of the Tombstone volcanic center [*Moore*, 1991] and are part of the caldera floor associated with eruption of the Uncle Sam Tuff [*Lipman and Sawyer*, 1985] and formation of the Tombstone caldera around 74 Ma [*Marvin et al.*, 1973]. Between the andesitic rocks and overlying Uncle Sam Tuff are two rhyolitic tuffs (AT1, AT2, Table 1) which are also part of the caldera floor. Tuff AT1 is commonly hydrothermally altered, and sericite from a cross cutting vein in the Charleston lead mine has a K-Ar age of 76.4 ± 3 Ma [*Reynolds et al.*, 1986]. Exposures of tuff AT2 in the western part of the district show it in normal fault contact with the overlying Uncle Sam Tuff, and these exposures are interpreted as part of the eroded caldera rim [*Lipman and Sawyer*, 1985]. The thickness of tuff AT2 and the large size and abundance of lithic fragments within it suggest that this unit might also have come from the Tombstone volcanic center [*Moore*, 1991].

The main mass of Uncle Sam Tuff in the Tombstone district is exposed over an area of ~15 by 20 km and is interpreted as little-deformed intracaldera fill, with associated collapse breccia, that is at least 1 km thick [*Lipman and Sawyer*, 1985]. The intracaldera and outflow facies of the Uncle Sam Tuff appear similar in outcrop and are moderately to densely welded. The Uncle Sam Tuff has a dacitic composition and a K-Ar biotite age of 73.5 ± 2.8 Ma [*Marvin et al.*, 1973]. The Schieffelin Granodiorite is postulated by *Lipman and Sawyer* [1985] to be a ring fault intrusion that intrudes the intracaldera Uncle Sam Tuff; the granodiorite has a K-Ar biotite age of 76 ± 3 Ma [*Marvin et al.*, 1973]. Younger basalt and andesite dikes with generally northeast trends intrude the Schieffelin Granodiorite and older rocks in the district and are Late Cretaceous(?) in age [*Moore*, 1993]. Postcaldera rhyolite and rhyodacite intrusions in the eastern part of the Tombstone district have K-Ar biotite, sanidine, and K-feldspar ages between ~64 and 67 Ma [*Reynolds et al.*, 1986].

Late Cretaceous and early Tertiary rocks in the northern Dos Cabezas Mountains (Figure 3) consist of dacitic pyroclastic breccias and basal and capping rhyodacitic welded tuff; these rocks are intruded by fine-grained granodiorite [*Drewes et al.*, 1988]. *Lipman and Sawyer* [1985] interpret the entire pyroclastic assemblage as a 3- to 4-km thick caldera fill sequence dipping homoclinally southwest, as indicated by compaction foliation within the welded tuff. Mineralized rock is associated with the Silver Camp quartz diorite [*Erickson*, 1981] which has a K-Ar biotite age of 64.3 ± 1.9 Ma.

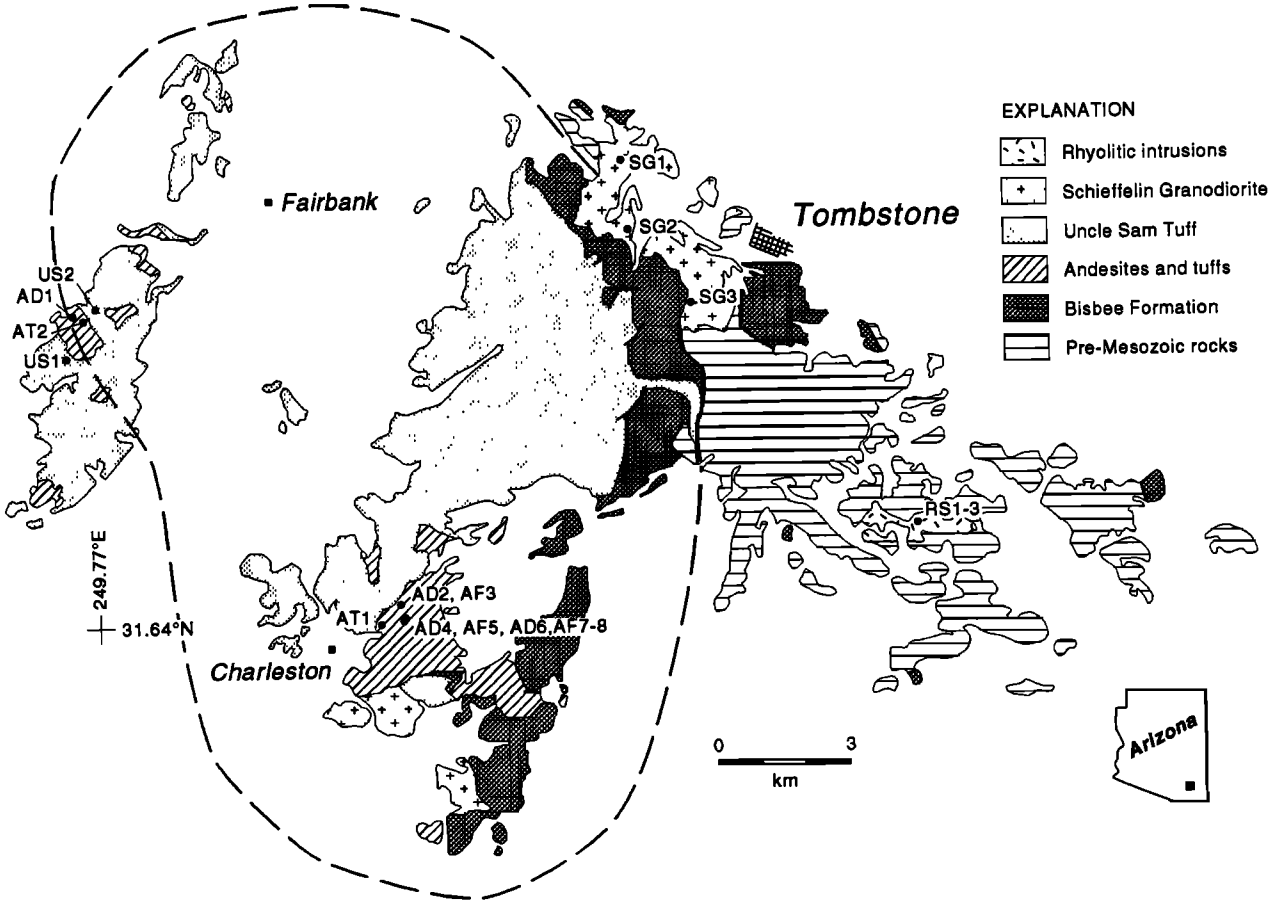


Figure 2. Geologic map of the Tombstone Hills [after *Gilluly*, 1956; *Drewes*, 1980] showing locations of the paleomagnetic sampling sites. Fine lines indicate contacts, thick lines indicate faults, and dashed line indicates the inferred margin of the Tombstone caldera [*Lipman and Sawyer*, 1985; *Moore*, 1993].

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Table 1. Paleomagnetic Data for the Tombstone District and Dos Cabezas Mountains, Arizona

	I_i	D_i	I_c	D_c	N	R	k	α_{95}	λ	ϕ
Tombstone Sites										
Rhyolite Sill (31.7°N, 250.0°E)										
RS1	65.1	340.1	63.1	335.5	5	4.9555	90	8.1	67.0	201.0
RS2	61.5	0.8	60.4	355.8	5	4.9569	93	8.0	79.8	232.0
RS3	54.6	357.1	53.3	353.4	5	3.9575	94	7.9	84.1	183.0
Schieffelin Granodiorite (31.7°N, 249.9°E)										
SG1	81.0	28.4	81.0	28.4	5	4.9792	192	5.5	46.6	253.9
SG2	72.6	28.8	72.6	28.8	5	4.8901	36	12.9	57.3	278.2
SG3	73.0	5.1	73.0	5.1	7	6.9862	436	2.9	62.9	255.7
Uncle Sam Tuff (31.7°N, 249.8°E)										
US1	77.6	42.8	76.4	8.0	8	7.8155	38	9.1	57.1	256.2
US2	81.3	341.1	74.2	323.5	8	7.9847	457	2.6	52.6	221.0
Ash Flow Tuffs										
AT1	66.5	30.6*	41.4	341.6	8	7.9561	159	4.4	(61.3	294.1)
AT2	67.9	328.1	60.4	322.1	9	8.9245	106	5.0	58.3	188.4
Andesite Dikes and Flows (31.7°N, 249.8°E)										
AD1	52.0	316.2*	12.1	313.9	12	11.7018	37	7.2	(53.2	173.4)
AD2	53.6	5.2*	23.5	342.1	7	6.7107	21	13.6	(85.0	309.4)
	-67.4	151.4*	-28.5	139.2	3	2.8776	16	31.5	(61.9	209.5)
AF3	84.5	205.0*	51.1	301.5	5	4.8464	26	15.3	(21.7	245.1)
AD6	56.0	346.7*	20.3	330.9	5	4.9388	65	9.5	(78.0	187.4)
	-53.0	194.8*	-26.6	167.5	2	1.9966	290	14.7	(77.4	327.5)
AF7[a]	-15.9	99.2	18.7	98.7	7	6.5044	12	18.1	-2.3	327.3
AF8[a]	66.4	149.0	71.0	286.5	6	5.6503	14	18.3	34.7	208.6
Overall Mean			67.8	351.7	(16)	15.4690	28	7.1	70.0	234.5
Dos Cabezas Sites										
Silver Camp Canyon Quartz Diorite (32.2°N, 250.4°E)										
SC1	64.5	5.3	64.5	5.3	5	4.9494	79	8.7	75.4	265.1
SC2	65.8	8.7	65.8	8.7	6	5.9435	88	7.2	72.9	270.5
SC3	66.1	10.9	66.1	10.9	6	5.9307	72	7.9	71.9	274.2
SC4	62.5	6.8	62.5	6.8	6	5.9097	55	9.1	77.2	273.1
SC5	55.3	25.0	55.3	25.0	6	5.9112	56	9.0	69.0	323.6
SC6	62.2	21.4	62.2	21.4	6	5.9568	116	6.3	69.8	300.5
Mean			63.0	13.7	(6)	5.9764	212	4.6	73.8	287.7
Intracaldera Welded Tuff (32.2°N, 250.3°E)										
WT1	24.3	357.1	44.5	331.8	8	7.9690	226	3.7	64.7	153.8
WT2	28.2	16.6	58.8	349.3	6	5.9818	274	4.1	78.7	203.6
WT3	18.0	1.7	42.4	341.9	7	6.9851	402	3.0	72.4	139.2
WT4	22.8	358.0	43.9	334.1	6	5.9668	151	5.5	66.5	150.9
WT5	29.2	335.0	33.8	309.4	6	5.9526	106	6.6	42.7	156.2
WT6[a]	18.0	312.8	11.7	301.9	3	2.9976	824	4.3	30.0	147.4
WT7[a]	28.6	303.1	13.9	287.9	5	4.9668	121	7.0	18.9	156.8

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Table 1. (continued)

	I_i	D_i	I_c	D_c	N	R	k	α_{95}	λ	ϕ
WT8	48.1	351.4	55.0	297.5	3	2.9817	110	11.8	38.9	182.3
WT9	24.4	1.1	47.0	335.6	6	5.9625	133	5.8	68.6	155.9
Mean			47.7	328.3	(7)	6.8070	31	11.0	62.6	161.5
Overall Mean			56.8	344.9	(13)	12.3736	19	9.7	77.8	195.3

I_i and D_i are in situ inclination and declination (in degrees) of paleomagnetic directions (I_c, D_c, corrected for structural tilt); N is number of samples (sites) analyzed; R is vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953]; α_{95} (A_{95} for pole) is the radius of 95% confidence (in degrees); and λ and ϕ are latitude and east longitude of virtual geomagnetic pole (in degrees) corrected (uncorrected) for structural tilt..

* In situ direction is used in overall mean; see text.

[a] Value is not included in overall mean; see text.

Methods

Paleomagnetic investigations of tectonic deformation and vertical axis rotations generally follow one of two approaches. (1) Rocks emplaced over a significant period of time (~10,000 years) are sampled in order to average out paleosecular variation (PSV) of the ancient geomagnetic field, and mean paleomagnetic poles (PP, Table 2) are compared with reference

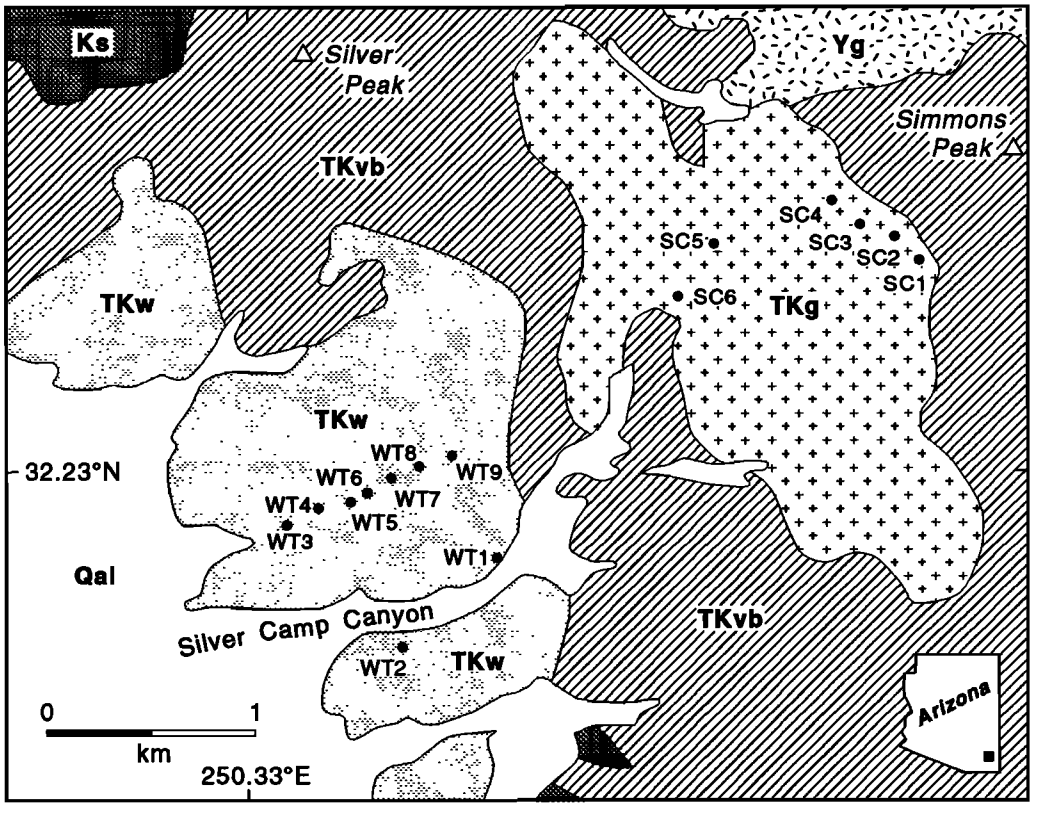


Figure 3. Geologic map of part of the Simmons Peak quadrangle in the Dos Cabezas Mountains [after *Drewes*, 1986] showing locations of the paleomagnetic sampling sites (WT1-9, intracaldera welded tuff; SC1-6, Silver Camp Canyon granodiorite). Map unit symbols are Qal, alluvium (Quater-nary); TKg, quartz diorite (Late Cretaceous or early Tertiary); TKw, rhyodacite or quartz latite welded tuff (Late Cretaceous or early Tertiary); TKvb, rhyolitic and dacite breccia (Late Cretaceous or early Tertiary); Ks, sedimentary rocks (Cretaceous); and Yg, granodiorite and granite (Middle Proterozoic).

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Table 2. Poleward Displacement and Vertical Axis Rotation of Localities in Southeastern Arizona

	λ	ϕ	A_{95}	N	d	r	Reference
Dos Cabezas Mountains (32.2°N, 250.4°E)							
N.A.-K	82	210	7	(26)			1
PP	78	195	13	(13)	0 ± 12	-6 ± 15	2
Gunnison Hills (32.1°N, 250.0°E)							
N.A.-C	32	127	7	(7)			3
PP	38	132	10	9	7 ± 10	2 ± 10	4
Tombstone Hills (31.7°N, 249.9°E)							
N.A.-K	82	210	7	(26)			1
PP	70	235	11	(16)	13 ± 10	-2 ± 15	2
Naco Hills (31.4°N, 250.0°E)							
N.A.-C	32	127	7	(7)			3
PP	37	140	6	62	9 ± 7	9 ± 8	4
Whetstone Mountains (31.8°N, 249.6°E)							
N.A.-C	32	127	7	(7)			3
PP	39	147	8	11	13 ± 8	13 ± 9	4
Mustang Mountains (31.7°N, 249.5°E)							
N.A.-Jr	61	142	7	(3)			5
VGP(?)	53	156	5	6	-3 ± 7	-11 ± 7	6
Tucson Mountains (32.2°N, 248.9°E)							
N.A.-K	82	210	7	(26)			1
PP	86	262	8	(26)	2 ± 8	8 ± 10	8
Huachuca Mountains (31.3°N, 249.7°E)							
MM	53	156	5	6			6
VGP	69	115	5	10	7 ± 6	25 ± 6	6
Canelo Hills (31.5°N, 249.5°E)							
MM	53	156	5	6			6
VGP-CP	70	76	10	(5)	11 ± 9	38 ± 9	7
VGP-CC	75	60	7	(9)	6 ± 7	43 ± 7	7
VGP-PC	67	14	7	(14)	6 ± 7	60 ± 7	7
VGP-CH	77	40	5	14	3 ± 6	47 ± 6	6
Silver Bell Mountains (32.3°N, 248.5°E)							
N.A.-K	82	210	7	(26)			1
PP	71	310	8	(34)	-1 ± 8	28 ± 11	9

Abbreviations are same as in Table 1. Parameters d and r are poleward displacement and clockwise rotation at locality with 95% confidence, respectively. N.A. is North American craton; K, Upper Cretaceous; C, Mid-Carboniferous; Jr, Upper Jurassic; PP, paleomagnetic pole; VGP, virtual geomagnetic pole (see text); CP, Canelo Pass; CC, Collins Canyon; PC, Parker Canyon; CH, sites CU1 and CU2. References are 1, *Gunderson and Sheriff* [1991]; 2, this study; 3, mean of Carboniferous poles (see Table 3) from *Van der Voo* [1989,Table 1]; 4, *Nick et al.* [1991]; 5, *Steiner and Helsley* [1975]; 6, *Hagstrum and Lipman* [1991a]; 7, *May and Butler* [1987]; 8, *Hagstrum and Lipman* [1991b]; and 9, *Hagstrum and Sawyer* [1989].

paleopoles to determine absolute vertical axis rotations with respect to the North American craton [e.g., *Hagstrum and Sawyer*, 1989]. (2) A single widespread volcanic unit (e.g., ash flow sheet), assumed to have cooled quickly with respect to PSV, is sampled, and declinations (or virtual geomagnetic poles (VGP), Table 2) are compared directly to determine relative vertical axis rotations between localities [e.g., *Hagstrum and Lipman*, 1991b]. In this study, samples were collected from localities in a number of different rock units in the Tombstone district and the Dos Cabezas Mountains to average out PSV.

Oriented samples for paleomagnetic analysis were collected at 18 sites in the Tombstone Hills (Figure 2) using both magnetic and solar compasses. Areas of hydrothermal alteration were generally avoided, and only the freshest possible exposures were collected. Attitudes for the extrusive rock units are based on flow contacts or interbedded sediments where available and on compaction foliations for pyroclastic rocks. Structural corrections for intrusive rocks (i.e., andesite dikes and rhyolite sill) were determined from the attitudes of stratified host rocks either at the site or nearby. No direct attitude could be obtained for the Schieffelin Granodiorite, but it is inferred to be untilted because it likely intrudes basically undeformed Uncle Sam Tuff, its top contact is flat lying, and fold-axis planes in its host rocks are vertical. Similarly, paleomagnetic samples were collected at 15 sites in the northern Dos Cabezas Mountains in the vicinity of Silver Camp Canyon (Figure 3). Nine sites were collected across a steeply dipping section of intracaldera welded tuff, and six sites were collected from stream wash exposures of the younger granodiorite up Silver Camp Canyon. No independent attitude is available for these intrusive rocks.

The remanent magnetization of individual specimens, cut from the core samples, was measured with a cryogenic magnetometer. Specimens from all samples were progressively demagnetized with either standard alternating field (af) or thermal procedures. Lines representing the characteristic magnetization directions were fitted to the demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980], and the statistics of *Fisher* [1953] are used in analyzing the site, unit, and overall mean directions and poles (Table 1). Estimates of poleward displacement and vertical axis rotation and their 95% confidence limits (Table 2) are calculated using the methods of *Beck* [1980] as modified by *Demarest* [1983].

Results

In general, natural remanent magnetizations (NRM) measured prior to demagnetization treatments are more variable in direction and intensity for samples from natural exposures than for samples from artificial exposures. The dispersion in NRM directions therefore is most likely due to isothermal remanent magnetizations caused by lightning strikes. In addition, some samples show NRM directions near the present-day field direction in southeastern Arizona that are probably due to recently acquired thermoviscous magnetizations. Af demagnetization was usually successful in removing these overprint components and in isolating the characteristic remanent magnetizations of the rock samples. On the basis of thermal and af demagnetization behavior, magnetite is presumed to be the dominant, if not sole, carrier of the remanence in most rock types. Representative demagnetization diagrams for specimens of each rock type from the Tombstone Hills and Dos Cabezas Mountains are given in Figures 4 and 5, respectively.

More complex demagnetization behavior, however, was also observed in some specimens of andesite dikes and flows in the southern Tombstone Hills. Usually ~20 to 30%, and sometimes more, of the NRM intensity remained after the maximum 100-mT af demagnetization treatment. In some cases, as much as 80% of the NRM intensity remained after the 579°C heating step. These results imply that hematite is present in varying

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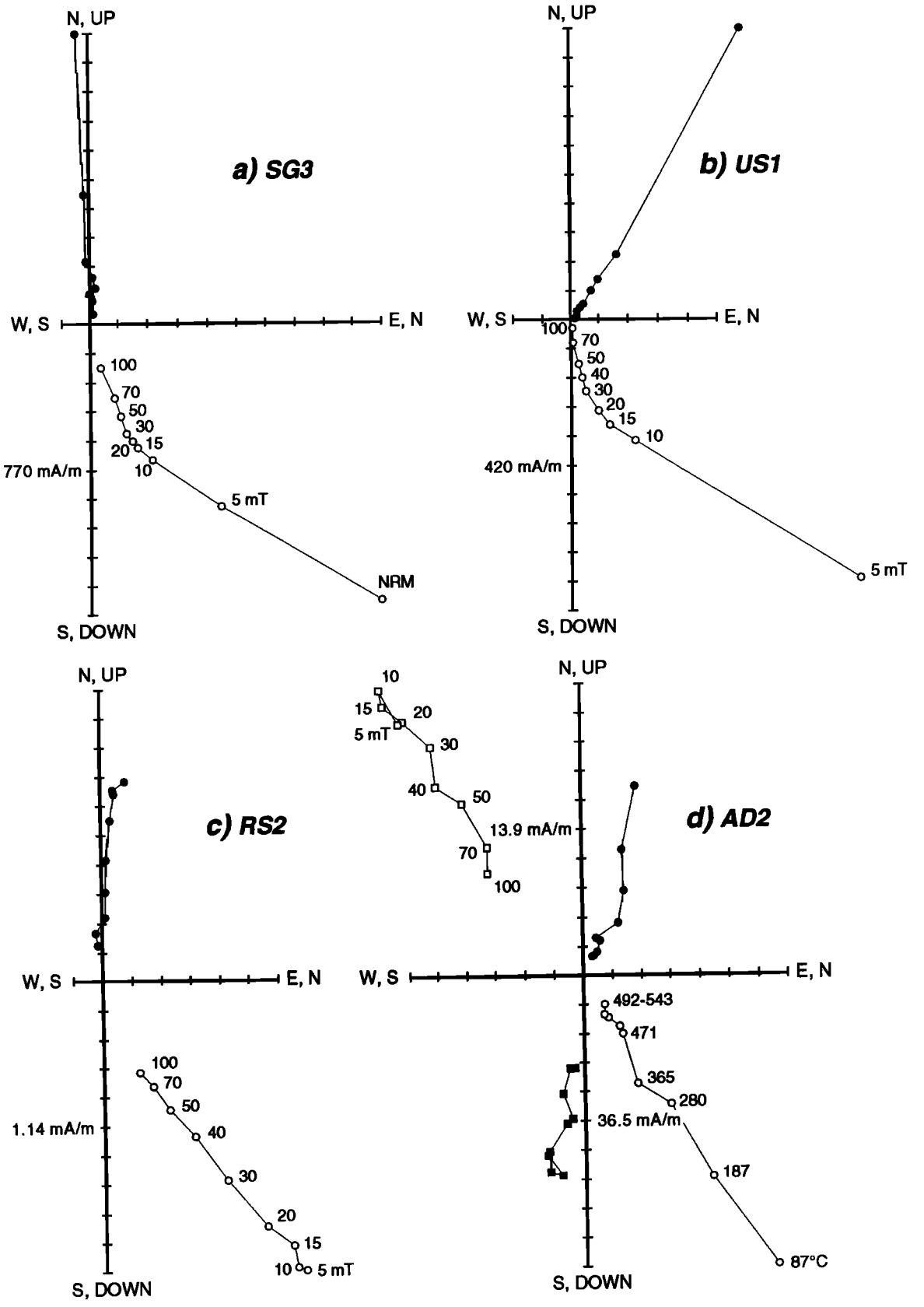


Figure 4. Orthogonal projections of in situ alternating field (af) (in milliteslas) and thermal (in degrees Celsius) demagnetization data for specimens of (a) Schieffellin Granodiorite from site SG3, (b) Uncle Sam Tuff at site US1, (c) rhyolite sill at site RS2, and (d) andesite dike (two specimens of normal and reversed polarity) at site AD2 in the Tombstone Hills. Solid (open) symbols indicate projections of vector endpoints onto the horizontal (vertical) plane.

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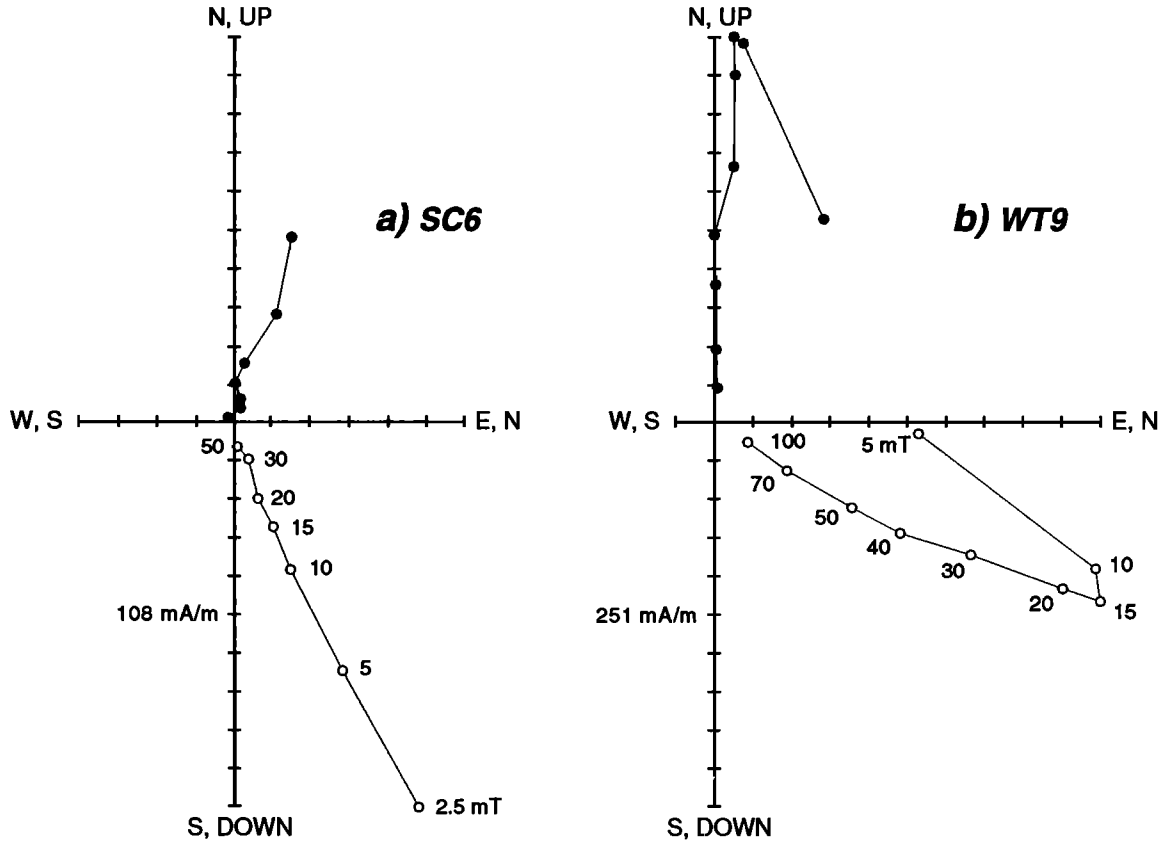


Figure 5. Orthogonal projections of in situ af demagnetization data for samples of (a) the Silver Camp quartz diorite from site SC6, and (b) welded tuff at site WT9 in the Dos Cabezas Mountains. Solid (open) symbols indicate projections onto the horizontal (vertical) plane.

amounts within these rocks and that it can be the dominant magnetic carrier. Specimens having both normal and reversed polarity components are found at sites AD2 and AD6. Multiple components of magnetization are also observed within individual andesite specimens; commonly, these components are superimposed and appear in demagnetization vector diagrams as sweeping arcs, rather than as distinct line segments. No consistent characteristic component of magnetization could be isolated in specimens from sites AF4 and AF5, and they are therefore absent from Table 1. Consistent within-site directions are found in specimens from sites AF7 and AF8, but the site-mean directions are clearly divergent, possibly due to overlapping normal and reversed polarity components, and have been omitted from the overall mean.

Site-mean directions for the Tombstone Hills are shown in Figure 6. Each site is assumed to represent an independent record of the ancient geomagnetic field. Site US1 is in the outflow facies of the Uncle Sam Tuff, and site US2 is in intracaldera tuff, and these sites evidently cooled and acquired their magnetizations at different times. Although the magnetic mineralogy and demagnetization behavior of the andesite dikes and flows imply magnetization over an extended period of time (at least one reversal of the geomagnetic field), the alteration is most likely related to emplacement of either the Schieffelin Granodiorite or the dikes themselves. The andesite flows are the most altered rocks in the area and commonly host ore deposits; alteration in the nearby Charleston lead mine has a similar age (~76 Ma) to that of the Schieffelin Granodiorite [see *Moore*, 1993]. Remagnetization of the andesite flows therefore most likely occurred after structural tilting of these rocks (220°E, 40°NW). In addition, some of the andesite dikes in this area are likely younger than the granodiorite and are probably untilted as well. The overall mean for the Tombstone Hills (Tables 1 and 2) likely averages PSV and is similar to the expected Late Cretaceous direction for stable North America at that locality [*Gunderson and Sheriff*, 1991].

Site-mean directions for the Dos Cabezas Mountains (Figure 7) apparently show a wide range of PSV. Sites SC1 through SC4 near the eastern margin of the Silver Camp quartz diorite probably cooled quickly and by themselves represent little, if any, PSV. Site SC6 at the western margin of the pluton has a direction similar to those of the eastern sites and probably cooled closer in time to the eastern sites than site SC5 near the center of the pluton. Similarly, sites near the base (WT3, WT4) and top (WT1, WT9) of the intracaldera tuff section have similar mean directions, whereas those in the middle, which most likely cooled later and over a longer period of time, have widely dispersed directions. Means for sites WT6 and WT7 are 54° and 49° away from the overall mean (2σ = 53°), respectively, and are interpreted as showing extreme PSV (excursion or transition); these directions have been omitted from the final mean calculation.

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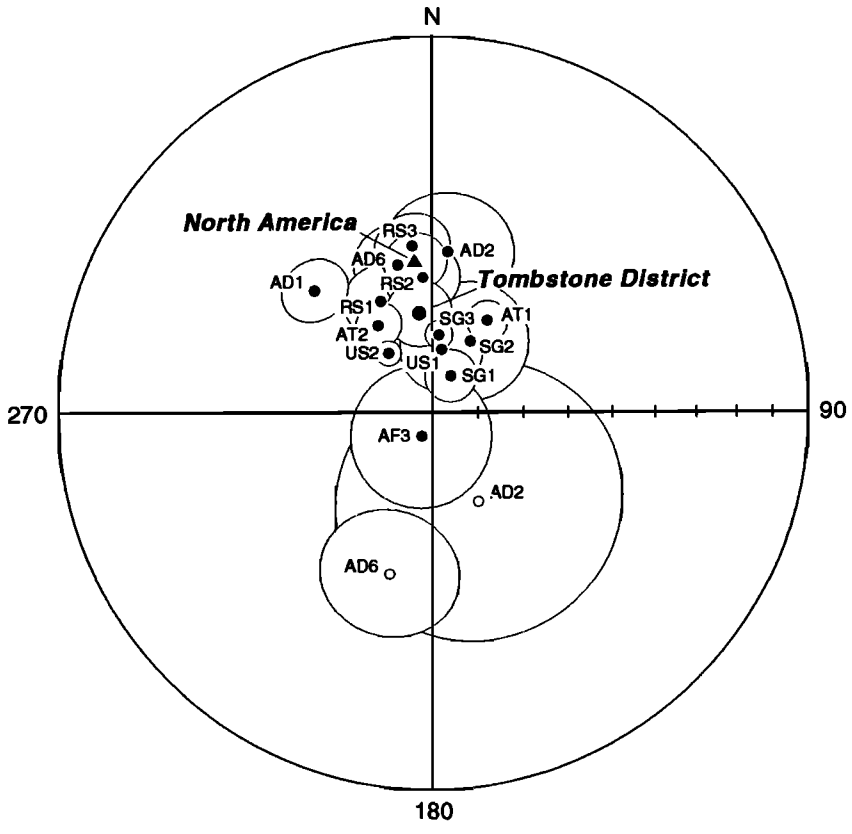


Figure 6. Mean directions with 95% confidence limits for sites in Late Cretaceous rocks of the Tombstone Hills (see Table 1). The large, solid circle indicates the overall mean direction, and the solid triangle indicates the expected Late Cretaceous direction for the locality based on the North American reference pole of *Gunderson and Sheriff* [1991]. Diagram is an equal-area plot in which solid (open) symbols indicate projection from the lower (upper) hemisphere.

The in situ paleomagnetic directions for the Silver Camp quartz diorite are in general agreement with the structurally corrected directions for the welded tuff (136°E, 40°SW). Tilting of the granodiorite therefore is presumed to be minor and probably no more than that associated with Basin and Range faulting and uplift of the Dos Cabezas Mountains (<10°-15°). Paleomagnetic data for resurgent plutons related to the Questa caldera in northern New Mexico [*Hagstrum and Lipman*, 1986] and the Tucson caldera in the Tucson Mountains [*Hagstrum and Lipman*, 1991a] show little subsequent tilting of these bodies even though compaction foliations of the host intracaldera tuff in both cases are nearly vertical. An unrecognized tilt of the granodiorite would increase the error associated with the overall mean but probably does not significantly affect its consistency with the expected Late Cretaceous direction for North America (Table 1).

Discussion

The similarity of mean paleomagnetic poles for rocks in the Tombstone Hills and the Dos Cabezas Mountains (presumed to average PSV) with North American reference poles (Figure 8a) implies that neither of these localities has undergone significant vertical axis rotations with respect to the stable interior since Late Cretaceous or early Tertiary time. A Late Cretaceous (~76-64 Ma) pole for the Tombstone Hills and Dos Cabezas Mountains, using the site means presented in Table 1, is 73°N, 224°E, concentration parameter (k) = 11, radius of 95% confidence (A_{95}) = 8° (Table 3). The concentration parameter is lower (scatter is greater) than would be expected for dispersion due to PSV alone [*McFadden et al.*, 1988], probably because of incomplete or incorrect structural corrections for some of these rocks. A Late Cretaceous (~71-81 Ma) paleopole for the Adel Mountains of west central Montana is 82°N, 210°E, k = 18, A_{95}= 7° [*Gunderson and Sheriff*, 1991], and a mid-Cretaceous (~85-129 Ma) paleopole calculated from data for four widespread localities is 71°N, 196°E, k = 352, A_{95}= 5° [*Globerman and Irving*, 1988]. The Cretaceous pole for southern Arizona is closer in latitude to the mid-Cretaceous pole (Figure 8a), implying that it might be nearer in age to the mid-Cretaceous pole and rotated clockwise 10° ± 10°. In contrast, a paleomagnetic pole for Late Cretaceous rocks in the Silver Bell Mountains to the west indicates ~30° of clockwise rotation [*Hagstrum and Sawyer*, 1989]; these and other available results for southeastern Arizona are presented in Table 2 and Figure 9.

The absence of significant vertical axis rotations in the Tombstone Hills (TH) and Dos Cabezas Mountains (DC) is generally consistent with paleomagnetic data for Pennsylvanian terra rosa paleosols of the Black Prince Limestone in

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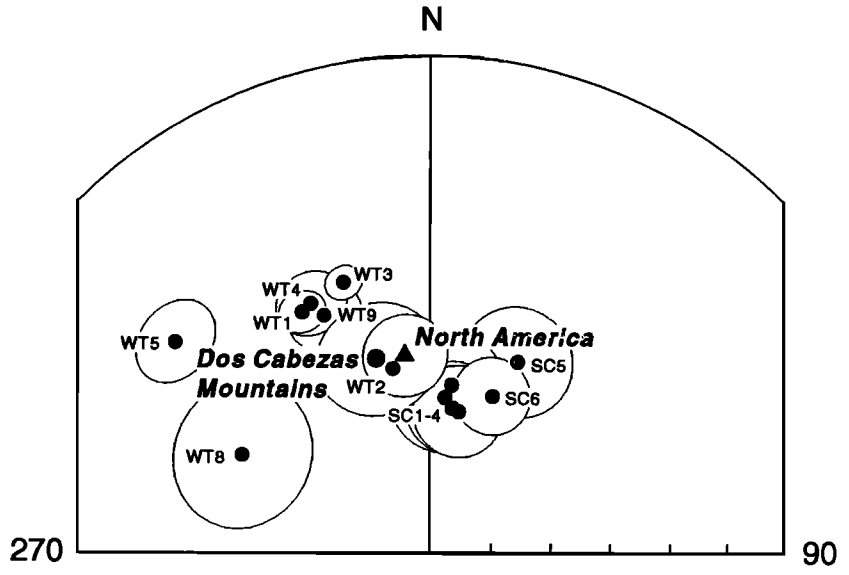


Figure 7. Equal-area plot of site-mean directions for the Dos Cabezas Mountains. Welded tuff means are corrected and quartz diorite means are not corrected for structural tilt (Table 1). Large solid circle and triangle indicate the overall mean of observed site-mean directions and the expected Late Cretaceous direction [*Gunderson and Sheriff*, 1991], respectively. Diagram is an equal-area plot, and data points are in the lower hemisphere.

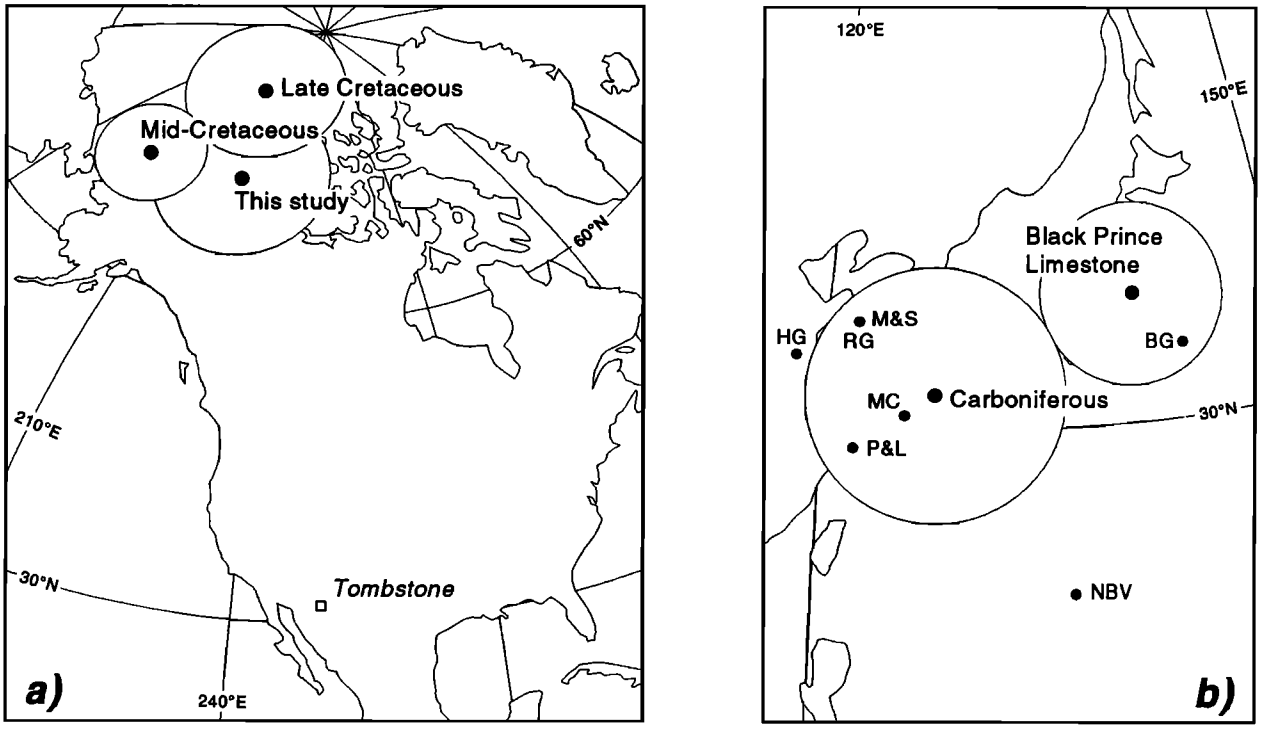


Figure 8. (a) North American reference paleopoles (with 95% confidence limits) for the mid-Cretaceous [*Globerman and Irving*, 1988] and Late Cretaceous [*Gunderson and Sheriff*, 1991], including the paleomagnetic pole and sampling site for this study (see Table 3). (b) North American Carboniferous poles [*Van der Voo*, 1989] and mean pole (larger open circle with 95% confidence limit), including the mean pole for terra rosa paleosols of the Black Prince Limestone (Morrowan) of southern Arizona (Table 3).

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Table 3. Carboniferous and Cretaceous Paleomagnetic Poles for North America

Locality	λ	ϕ	k	A_{95}	Ref.
Cretaceous					
Adel Mountains, Montana	82	210	18	7	1
Tombstone and Dos Cabezas	73	224	11	8	2
Mid-Cretaceous, North America	71	196	352	5	3
Carboniferous (Namurian-Westphalian)					
Black Prince paleosols	37	140	13	5	4
Riversdale Group	36	122	91	6	5
Pomquet and Lismore Formations	29	122			5
Barachois Group	34	143	107	7	5
New Brunswick Volcanics	21	135	49	10	5
Hopewell Group	34	118	32	7	5
Maringouin and Shepody	36	122	89	3	5
Mauch Chunk Formation	31	125	16	8	5
Mean	32	127	77	7	

Abbreviations are same as in Table 1. References (Ref.) are 1, *Gunderson and Sheriff* [1991]; 2, this study; 3, *Globerman and Irving* [1988]; 4, *Nick et al.* [1991]; and 5, *Van der Voo* [1989, Table 1].

the nearby Gunnison Hills (GH), Whetstone Mountains (WM), and Naco Hills (NH) [*Nick et al.*, 1991], although data for the Naco Hills and Whetstone Mountains show small but significant clockwise rotations. Overall, these data suggest that rocks in the vicinity of the Tombstone district have remained basically concordant with the North American craton. The paleomagnetic data for the Paleozoic paleosols, however, in part show small but significant clockwise rotations (Table 2), and a comparison of these averaged results with a reference Carboniferous pole (Table 3 and Figure 8b) indicates 9° ± 7° of clockwise rotation relative to North America. Clockwise rotation of the Colorado Plateau was proposed by *Hamilton* [1981] to account for Laramide compression to the north and Tertiary extension along its eastern margin, and recent determinations have indicated that the total rotation is between ~5° [*Bryan and Gordon*, 1990] and 14° [*Kent and Witte*, 1993]. Thus rocks within the "unrotated" domain of southeastern Arizona might have been affected by smaller clockwise rotations similar to those affecting the Colorado Plateau.

The virtual paleomagnetic pole (VGP) for the Mustang Mountains is from a single ash flow sheet [*Hagstrum and Lipman*, 1991b] which cooled quickly and probably does not average PSV. Upper Paleozoic rocks are also exposed in the Mustang Mountains (MM, Figure 9) and are most likely a southern continuation, with no apparent structural break, of those in the Whetstone Mountains (WM). Thus we infer that these rocks have remained basically stable with respect to the

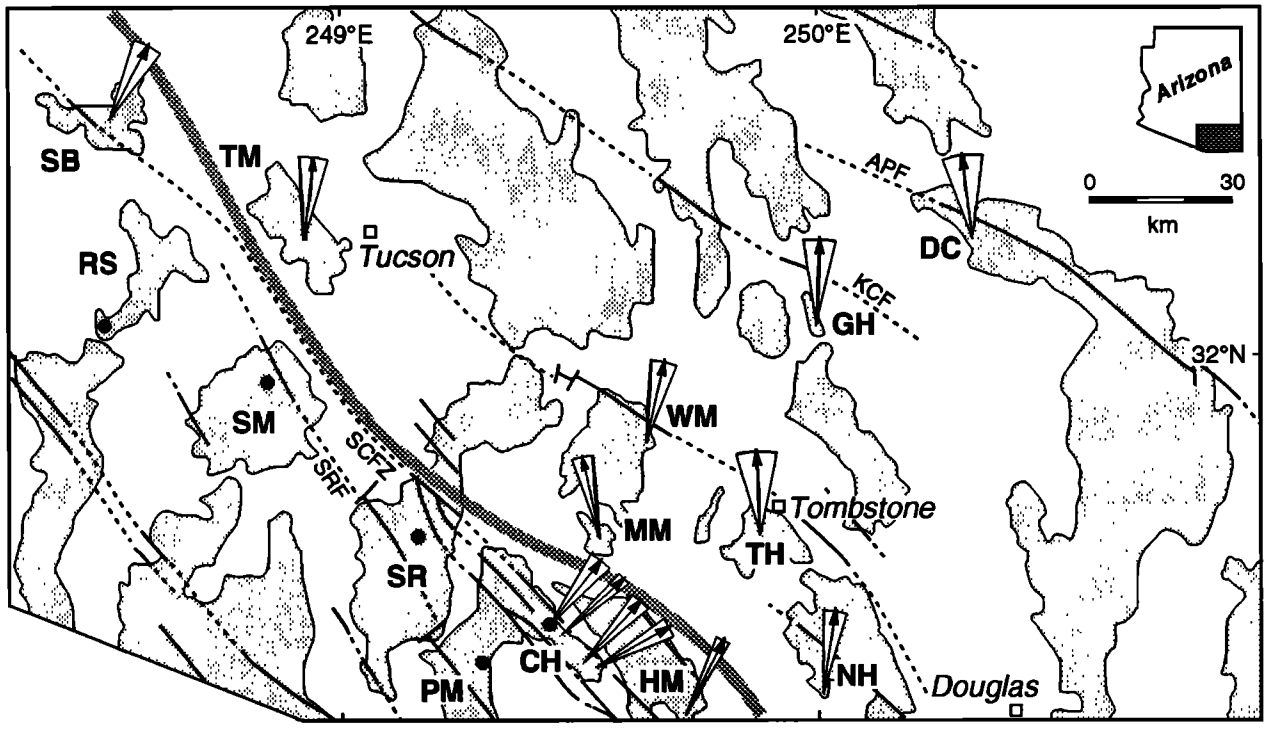


Figure 9. Regional map of southeastern Arizona showing vertical axis rotations relative to true north with their 95% confidence limits (Table 2). Dots indicate localities of paleomagnetic investigations for which rotations are unreliable or indeterminate (see text). Solid/dotted lines are regional northwest trending, high-angle faults [after *Drewes*, 1981]. Stippled, thick line is inferred rotational domain boundary which parallels the Sawmill Canyon fault zone (SCFZ) in the Santa Rita (SR), Canelo Hills (CH), and Huachuca Mountains (HM). Other abbreviations are same as in Figure 1; SRF, Santa Rita fault; KCF, Kelsey Canyon fault; and APF, Apache Pass fault.

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craton (although they may have been rotated along with the Colorado Plateau) and that the apparently significant counterclockwise rotation for the Mustang Mountains (-12° ± 7°, Table 2) is likely an artifact of PSV. The paleopole for this locality, however, serves as a fixed reference point for other localities within the same ash flow sheet. Results for such localities in the Canelo Hills (CH) and Huachuca Mountains (HM) to the south show large clockwise vertical axis rotations (~40°-60°) with respect to the Mustang Mountains [*Hagstrum and Lipman*, 1991b]. These data imply that a rotational domain boundary exists between the Mustang Mountains and the Canelo Hills and Huachuca Mountains (Figure 9).

Paleomagnetic data for the Upper Cretaceous Roskruge Volcanics [*Vugteveen et al.*, 1981] in the Roskruge Mountains (RM) have not been included in this analysis (Figure 9). Recent sampling of the Roskruge Volcanics near Bell Mountain [*Hagstrum et al.*, 1994] has shown that the number of samples (N) = 4 for the Upper Cretaceous pole of *Vugteveen et al.* [1981], rather than N = 9 as previously reported. This population of independent observations (site means) is probably too small to average out PSV. Also, a stable characteristic magnetization could not be isolated in 92 samples of the Paleocene Ruby Star Granodiorite from the Sierrita Mountains (J. Hagstrum, unpublished data, 1987). The Corral Canyon pole [*May et al.*, 1986] for the Patagonia Mountains (PM) has been omitted from our analysis as well because similar paleomagnetic results from the nearby Patagonia Granodiorite imply that this pole is based on a Paleocene remagnetization [*Hagstrum*, 1994]. In addition, rocks of the Glance Gonglomerate in the Canelo Hills [*Kluth et al.*, 1982] are within a caldera structure [*Lipman and Hagstrum*, 1992], have been affected by postemplacement potassium metasomatism [*Krebs and Ruiz*, 1987], and might have been remagnetized after deformation.

Remagnetization of the Glance Conglomerate may also imply remagnetization of the phenocryst-rich quartz rhyolite tuff (Canelo Hills Volcanics [*May and Butler*, 1987]) erupted from the Parker Canyon caldera [*Lipman and Hagstrum*, 1992]. *Hagstrum and Lipman* [1991b] used paleomagnetic directions for this tuff to determine regional vertical axis rotations in the Mustang Mountains, Canelo Hills, and Huachuca Mountains. Inclinations for the Canelo Hills Volcanics (and Glance Conglomerate), however, are shallower (~10°-20°) at Canelo Pass than at other localities in the region, and the Canelo Pass section is closest to the caldera margin; these rocks are therefore more likely to have been affected by potassium metasomatism localized along the caldera ring-fracture zone. Moreover, magnetizations in the Glance Conglomerate [*Kluth et al.*, 1982] are partly carried by hematite, which is often associated with areas of hydrothermal alteration and potassium metasomatism [*McIntosh*, 1991], whereas magnetite is the sole remanence carrier in the Canelo Hills Volcanics [*May and Butler*, 1987; *Hagstrum and Lipman*, 1991b]. Data for overprinted welded tuffs (in hematite) toward the base of the Mount Wrightson Formation [*Hagstrum*, 1994] in the Santa Rita Mountains (SR) indicate clockwise vertical axis rotation (~60°) similar to those observed in Jurassic tuffs in the Canelo Hills [*Hagstrum and Lipman*, 1991b], but N (2 or 3) is too small for this result to be taken seriously.

The northwest trending rotational domain boundary inferred for southeastern Arizona (Figure 9) is parallel to major complex (or multiphased) high-angle faults in the region [*Drewes*, 1981]. In particular, a part of this apparent boundary is parallel to the present trend of the Sawmill Canyon fault zone between the Mustang Mountains and the Canelo Hills and Huachuca Mountains. Northwest trending faults have been intermittently active in southern Arizona since the Precambrian when large amounts of both strike-slip and dip-slip displacements occurred along them [*Drewes*, 1981]. Most recently, these strike-slip faults were active during Laramide deformation and were important in the localization of intrusive bodies and as guides to the movements of mineralizing fluids [*Drewes*, 1981; *Keith and Wilt*, 1986].

For the most part, left-lateral displacements during the Laramide orogeny have been inferred for strike-slip faults in southern Arizona [*Drewes*, 1981], although evidence pertaining to these movements has been difficult to obtain because of poor exposures and the lack of features indicating the sense and amount of displacement. Clockwise vertical axis rotations in the region, however, might result from large-scale dextral shear, implying right-lateral displacement along the northwest trending strike-slip faults, some of which probably coincided with the rotational domain boundary. The clockwise rotations and right-lateral displacements in southeastern Arizona are possibly linked to regional dextral shear along the western margin of North America in Late Cretaceous and early Tertiary time caused by northward oblique subduction of oceanic plates relative to the continental interior [*Beck*, 1980; *Cox*, 1980; *Engebretson et al.*, 1984].

Conclusions

Paleomagnetic data presently available for Late Cretaceous and older rocks in southeastern Arizona, including new data for the Tombstone district and Dos Cabezas Mountains, are generally consistent with two rotational domains (rotated and unrotated) separated by a northwest trending boundary (Figure 9). This boundary roughly coincides with the exposed Sawmill Canyon fault zone in the Santa Rita Mountains, Canelo Hills, and Huachuca Mountains and is parallel to other northwest trending strike-slip faults in the region that were active during Laramide deformation.

The paleomagnetic data southwest of the boundary indicate clockwise rotations about vertical or near-vertical axes between ~25° and 60° relative to stable North America. Variation in the magnitude and sense of vertical axis rotation within the rotated domain to the southwest is unknown, and this domain is perhaps part of a larger pattern of clockwise rotation, developed since the mid-Cretaceous, observed along the western margin of North America [*Beck*, 1980, 1988]. Paleomagnetic data for the Baja California Peninsula imply that these outboard rocks have also been rotated ~30° clockwise (and translated ~10° northward) since the mid-Cretaceous [*Hagstrum et al.*, 1985]. The clockwise rotations in southern Arizona are consistent with a pattern of regional dextral shear (with no significant northward translations, Table 2), and the northwest trending strike-slip faults in the area probably accommodated right-lateral displacements during the Laramide orogeny.

Although paleomagnetic data for the unrotated domain

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generally indicate that these rocks have remained stable relative to North America since Paleozoic time, data for the Carboniferous paleosols apparently indicate a relatively minor clockwise rotation of 9° ± 7°. The Late Cretaceous paleopole for the Tombstone district and Dos Cabezas Mountains (Table 3 and Figure 8a) therefore likely can not be considered a high-quality reference paleopole for cratonic North America. This rotation is similar to the cumulative clockwise rotation of the Colorado Plateau (~5°-13°) since the mid-Cretaceous, and rocks of the unrotated domain might have undergone lesser rotations similar to those affecting the Colorado Plateau.

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